// Mohamed Elshaarawy
cairo, Egypt
+201032883092 mohamedelshaararwy@gmail.com mohamedelshaarawyy/ Mohamed-Shaarawy

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Education

University of Prince Edward Island **October 2021 - Present**
Bachelor of Science in Computer Science with a specialization in Video Game Programming

- Relevant coursework: Data Structures and Algorithms, Database, Computer Graphics , Video Game Design, Advanced Graphics, Software Design and Architecture, Game Engine Architecture, Design and Analysis of Algorithms, Data Science, Machine Learning.

Highlights of Qualifications

- I am a self-motivated and quick learner with a strong ability to adapt to modern technologies, concepts, and dynamic situations.
- Possess excellent analytical and problem-solving skills, coupled with effective communication, interpersonal, and team work abilities.
- Fluent in both English and Arabic.

Technical Skills

- **Languages**: Python, Java, C++, C#, C, SQL, Kotlin, bash, TypeScript.
- **Frameworks**: OpenGL, QT.
- **Tools**: Unity, Unreal Engine 5, shaders, Maya, Git/GitHub/Gitlab/perforce, Android Studio, Firebase, Docker, GCP
- **Techniques**: Design patterns, Version Control, Shaders,Scene Optimization, Game Design Document, Game Engine Architecture.

Projects

Solar System Simulation | *OpenGL, C++* **April 2024 - May 2024**

- Developed a 3D spaceship game simulating solar system navigation with rotating planets and debris.
- Implemented a survival-based scoring system.

Game Design Document (GDD) **Nov 2024 - December 2024**

- Created a comprehensive Game Design Document for an exploration/adventure game with survival mechanics, story progression, and unique supernatural abilities.
- Designed engaging gameplay systems with chapter-based power upgrades, resource management, and crew role assignments.
- Brought the game world to life with diverse environments, dynamic weather, and interactive puzzles that encourage players to explore.
- Developed a functional game prototype in Unity using C# for concept validation.
- Market research and monetization strategy development: combined one-time purchase with optional cosmetic add-ons.
- GDD Summary

Library System | *Qt, C++* **oct 2024 - Dec 2024**

- Built a GUI-based library system, integrating user authentication, menu browsing, and order management
- Enabled dynamic price display and account balance management.

Library System with Database | *Java, MySQL, JavaFX* **oct 2024 - Dec 2024**

- Developed a fully functional library management system using Java, MySQL, and JavaFX.
- Implemented features for user authentication, book borrowing management, and database interaction.

Scene Optimization | *Unity* **November 2024 - December 2024**

- Optimized Unity scenes to enhance performance and frame rates.
- Applied various optimization techniques, including:
- - Baking Light for static lighting to reduce real-time rendering load.
- - Dynamic Batching to optimize rendering efficiency.

- - Level of Detail (LOD) Techniques to reduce GPU workload.
- - Occlusion Culling to enhance performance by preventing rendering of unseen objects.

Fighting Game Prototype | *Unreal Engine 5, C++, UE Blueprints* **Mar 2025 - May 2025**

- Developed a fighting game prototype with character movement, combat mechanics, and AI behavior.
- Implemented character animations and special moves using Unreal Engine's animation system.
- Created a user-friendly interface for game settings and Pause Menu.

Arkanoid Game | *Raylib, C++* **Apr 2025 - May 2025**

- Developed a 2D Arkanoid game, implementing ball physics, paddle control, and a scoring and health system.
- Optimized for responsiveness across multiple screen sizes.

News App | *Kotlin, RetroFit* **Apr 2025 - May 2025**

- Developed a news app with Kotlin and integrated API for browsing headlines, searching, filtering, and managing favorites.
- Ensured a responsive design suitable for various screen sizes.

Nearby Explorer App | *Kotlin, RetroFit* **Apr 2025 - May 2025**

- Built a nearby places app, integrated with Overpass API for user location tracking and POI discovery.
- Developed a responsive UI adaptable to different screen sizes.

Echos of You (Graduation Project) | *Unity, C#, Shader Graph, VFX Graph* **Oct 2025 - February 2026**
Team Lead, Gameplay Programmer, Art

- Lead a team in the development of Echoes of You, a mobile story-driven puzzle game with character abilities, puzzles, and combat mechanics.
- Designed and implemented character camera systems, authored custom shaders, and converted environment assets from Unreal Engine to Unity.
- Optimized texture sets by separating AORM (Ambient Occlusion, Roughness, Metalness) maps into individual channels for seamless integration into Unity's material workflow.
- Developed VFX systems and Shader Graph effects to enhance the game's atmospheric and visual storytelling.

Mirage (Graduation Project) | *Unreal Engine, C++, Blueprints, Materials, Control Rig, Sequencer***February 2026 - Present**
Team Lead, Gameplay Programmer, Art, Animation

- Leading development of a tactical stealth PC game featuring motion matching, traversal, and AI interactions.
- Integrated a dynamic cover system with wall rotation, auto-crouch, and smooth player transitions.
- Built reusable gameplay systems in C++ including pickup mechanics, throwable grenades, and sliding movement.
- Improved gameplay quality by debugging animation systems using Unreal Rewind Debugger.
- Created gameplay-driven animations using Control Rig and Sequencer for enemy reactions, status effects, and combat actions including EMP stun, smoke coughing, and grenade throwing.
- Optimized game performance by converting assets to Nanite, reducing draw calls, applying culling systems, and lowering lighting overhead.
- Managed team workflow, task assignments, and Perforce version control pipeline.